Theoretical Verification of Ohm's Law

Example 1: Let V=100 V, R=10Ω,Find the value of I. Assume that the resistance value is kept constant.

$$I = \frac{V}{R} = \frac{100}{10} = 10A$$

Example 2: Let V=10 V, R=10Ω, Find the value of I. Assume that the resistance value is kept constant.

$$I = \frac{V}{R} = \frac{10}{10} = 1A$$

From these two examples, we can conclude that the voltage increases (100 V) the current increases (10 A), and voltage decreases (10 V) current decreases (1 A). That is, current I is directly proportional to voltage V, proved theoretically.

Example 3: Let V=100 V, R=100 Ω, find the value of I.

$$I = \frac{V}{R} = \frac{100}{100} = 1A$$

Comparing Examples 1 and 3, in both the examples the voltage is kept constant and the resistance is varied. From these, it is proved that the resistance value decreases (10Ω), current increases (10A), whereas resistance value increases (100Ω), current decreases(1 A), As a result, we prove that I is inversely proportional to R.

Example 4: Let I=2 A, R=10Ω, find the value of V.

To find the value of V, the respective quantities are substituted in Ohm's law 2.

$$V = I \cdot R = 2 \times 10 = 20V$$

Example 5: Let I = 5 A, V=25 V, find the value of R.

Using Ohm's Law 3: $R = \frac{V}{I} = \frac{25}{5} = 5\Omega$

1.4.1 Power

The Electrical work done during a specific time interval is known as power.

$$Power(P) = \frac{Work\ done}{Time} = \frac{W}{T}$$

$$P = \frac{V \times I \times t}{t}$$

$$P = V \times I \quad or\ Power = Voltage \times Current$$

Power (P) is measured using the unit called watt. To represent high power values, kilo-watt or mega-watt is used.



The unit of Power is 'Watt'., named after Scotland scientist James Watt.

1 kilowatt = 1000 watt
i.e. 1 kw = 1,000 watt
1 Mw = 10,00,000 watt

To find and measure the power, the following formula is used

$$P = V \times I \quad \ldots\ldots\ldots\ldots\ldots\ 1$$

According to ohm's law, ($V = I\ R$). Substitute this in the equation..1

$$P = I \times R \times I = I^2 R \quad \ldots\ldots\ldots\ldots\ldots 2$$

And also substitute $I = \frac{V}{R}$ in equation 1,

we get $P = V \times \frac{V}{R} = \frac{V^2}{R}$

Therefore, power can be calculated by following three formulae.

1. $P = V \times I$
2. $P = I^2 R$
3. $P = \frac{V^2}{R}$

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